

Meeting Q&A Card — Four Ideas We Tested

For deck snapshot: `saq-meeting-summary@9c21469`.

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0.1 Decision In One Sentence

Close the four method lines, keep the baselines and negative evidence, and require the next direction to pass a cheap cross-baseline limitation test before substantial implementation.

0.2 Four Attempts In One Line Each

- **Attempt 1:** a small GIST estimator improvement did not replicate on CIFAR; the favorable lossy-projection test also ranked worse than native SAQ.
- **Attempt 2:** exact scalar training improved reconstruction more reliably than it improved search; its inner optimizer is prior work (Wu 1991; Grønlund et al. 2017).
- **Attempt 3:** distance quality changed one GIST interpretation, but not the method; 39.6% of paired queries worsened and DEEP remained negative.
- **Attempt 4:** the mathematical opportunity is real, but the registered exact pipeline exceeded its cost ceiling before real-data evaluation; later recovery produced no valid scientific result.

0.3 Likely Questions

All four stopped. What did we actually gain?

We ruled out four plausible paths under fixed decision rules and retained a stronger baseline, better evaluation practice, and earlier cost checks. The result is a smaller and better-justified search space, not four universal impossibility claims.

Why not continue Attempt 1 with another dataset, dimension, or rotation?

The registered positive effect failed its external replication, and the lossy version failed a deliberately favorable gate before quantization. Changing datasets, dimensions, or rules now would be a new study, not completion of the failed one.

Attempt 2 reduced reconstruction error. Why is that not enough?

Nearest-neighbor ranking depends on how errors interact with each query near the ranking boundary, not only on average vector reconstruction. The DEEP counterexample reduced raw reconstruction error by about 20% while Recall became worse.

Attempt 3 was 1.078 times faster. Why is that not the contribution?

It is one GIST operating point under a prior-work metric, with no new mechanism and no positive DEEP replication. The aggregate improved, but 39.6% of paired queries became worse.

Attempt 4 exceeded 24 hours by only about ten minutes. Why enforce the stop?

The 24-hour ceiling was fixed before observing the result; relaxing it afterward would make the gate meaningless. More importantly, 24.17 hours is a projection for the frozen construction-and-evidence pipeline, not a nearly complete real-data search result.

Does Attempt 4 prove arbitrary cardinalities are too expensive or do not help?

No. It proves only that the registered exact pipeline missed its affordability gate. No benchmark-data, Recall, throughput, or end-to-end systems result was produced.

Why classify the later Attempt 4 run as unknown instead of failed?

The run recorded a start but did not produce a trustworthy terminal record. Calling it a scientific failure would claim evidence that does not exist; the project decision is simply to stop spending effort on artifact recovery.

Why not present the bug fixes, exact solvers, or protocol work as the contribution?

They validate or govern experiments; they do not improve the database method's quality-speed frontier. Reviewers would correctly treat them as supporting work rather than scientific novelty.

What exactly should happen next?

Read the closest primary work, identify a limitation visible on at least two strong independent baselines, and propose one cheap frozen test with an early stop threshold. Do not implement a new method until that question and its cost boundary are approved.

0.4 Numbers Worth Remembering

- Attempt 1 GIST estimator error: about -0.60%; CIFAR replication failed.
- Lossy projection top-100 agreement: 0.99289 versus native SAQ 0.99462.
- Attempt 2 audio reconstruction improvement: about 15.5%; Recall gain was small and not stable across settings.
- Attempt 3 measured GIST speed ratio: 1.07793; paired queries worse: 39.6%.
- Attempt 4 projected cost: 24.17025 CPU-hours; frozen ceiling: 24.0.

0.5 Phrases To Avoid

- Do not say “PCA is optimal.” Say “the tested replacements did not justify continuation.”
- Do not say “exact training does not help.” Say “its search effect was not stable.”
- Do not say “Attempt 4 failed on real data.” Say “the registered pipeline stopped before real-data evaluation.”
- Do not say “we have no result.” Say “we have negative selection evidence, but no new method claim.”